Ahmedabad University

ECE 501: Digital Image Processing

Week 5 Progress Report

Group 6

Vehicle Orientation Estimation Using Conventional Image Processing

Submitted to faculty: Prof. Mehul Raval

Date of Submission: 17/10/2026

Student Details

Roll No.	Name of the Student	Email Id
AU2440098	Dhruvika Mehta	dhruvika.m@ahduni.edu.in
AU2440103	Krisha Desai	krisha.d@ahduni.edu.in
AU2440276	Manthan Pancholi	manthan.p1@ahduni.edu.in
AU2440262	Rim Shah	rim.s@ahduni.edu.in

GitHub Repository (DRASHTI-HaOBB):

https://github.com/ERYAGNIK003/DRASHTI-Dataset

Dataset Source:

DRASHTI-HaOBB – Heading-angle Oriented Bounding Box Dataset

https://zenodo.org/records/18278989

1. Work Done This Week

In Week 5, we focused on investigating the comparatively high angular error produced by the Hough Transform during the previous orientation-estimation experiment. The Week 4 results showed that Hough Transform had a higher mean angular error than PCA and contour-based estimation. Therefore, the objective of this week was to examine its performance in greater detail and identify the vehicle classes and individual samples associated with larger errors.

Instead of repeating the dataset extraction or running all three methods again, we used the existing Week 3 results containing the ground-truth orientations and estimated orientations for the selected 100 vehicle samples. The analysis focused exclusively on Hough Transform.

First, the angular errors were recalculated from the ground-truth and predicted orientations. A class-wise analysis was then performed to calculate the number of samples, mean error, median error, maximum error, and standard deviation for each vehicle class. The classes were ranked according to their mean angular error to identify those that presented greater difficulty for the Hough Transform.

Next, we analysed the relationship between ground-truth orientation and angular error. The orientations were grouped into ranges to compare the mean error across different angular intervals. Individual vehicle samples with the largest angular errors were also identified and saved for further investigation.

Six graphs were prepared to visualise the results. These included class-wise mean angular error, class-wise error distribution, ground-truth versus predicted orientation, error versus ground-truth orientation, orientation-range-wise mean error, and the overall error distribution.

The corresponding statistical tables and summary information were saved in the Week5_Results folder in Google Drive.

The purpose of this analysis was to understand where the Hough Transform performed poorly and investigate possible reasons for its higher error. The numerical analysis provides evidence of the error patterns, while further visual inspection of the detected Hough lines is required to confirm the specific causes.

The Week 5 workflow was:

Week 3 Results CSV → Hough Error Calculation → Class-Wise Analysis → Orientation-Wise Analysis → Identification of High-Error Samples → Graphs and Tables → Saving Results to Google Drive.

2. Methodology

2.1 Loading and Preparing the Results

The saved Week 3 CSV file was loaded into Python using the Pandas library. It contained the vehicle identifiers, vehicle classes, ground-truth orientations, and estimated orientations produced by Hough Transform.

The angular error was calculated using the difference between the ground-truth and estimated orientations:

$$Angular\ Error = |\theta_{predicted} - \theta_{ground\ truth}|$$

Both orientations were represented in the required 0°–90° range. Samples without a valid Hough prediction were treated as failed predictions and excluded from the mean-error calculation rather than being assigned an arbitrary error value.

The resulting dataset was used for all subsequent Week 5 analyses.

2.2 Class-Wise Angular Error Analysis

The valid predictions were grouped according to vehicle class. For each class, the following measures were calculated:

- Sample count
- Mean angular error
- Median angular error
- Maximum angular error
- Standard deviation of angular error

The classes were ranked in descending order of mean angular error. This helped identify which vehicle categories showed the largest errors in the selected sample.

A bar graph was generated to compare the mean angular error across the represented classes. A boxplot was also generated to examine the spread, median, and potential outliers within each class.

The number of samples in each class was considered when interpreting the results because the 100-image sample was not equally distributed across the vehicle categories.

2.3 Ground-Truth versus Predicted Orientation

A scatter plot was generated to compare the ground-truth orientations with the orientations estimated by Hough Transform.

The ground-truth orientation was plotted on the horizontal axis, while the estimated orientation was plotted on the vertical axis. A diagonal reference line represented ideal estimation, where the predicted orientation is equal to the ground truth.

The further away from this line the points are, the greater the difference between the estimated and actual orientations. The plot was used to determine samples where the Hough Transform gave incorrect predictions.

2.4 Error Analysis Across Orientation Ranges

To investigate whether estimation errors varied with vehicle orientation, the ground-truth orientations were divided into six ranges:

- 0°–15°
- 15°–30°
- 30°–45°
- 45°–60°
- 60°–75°
- 75°–90°

For each range, the number of samples and mean angular error were calculated. A scatter plot was also generated to show the relationship between the individual angular errors and their corresponding ground-truth orientations.

This analysis can be used to determine if there are specific ranges of orientation that are more difficult to estimate. Any relationship observed, however, should be considered in conjunction with the number of samples in each range.

2.5 Identification of High-Error Samples

The Hough predictions that were valid were sorted in descending order of angular error. Samples that yielded the highest errors were placed in a separate table with their identifiers, vehicle classes, ground-truth orientations, estimated orientations and angular errors.

These samples provide useful starting points for examining why the Hough Transform failed to estimate the vehicle's orientation accurately.

An error-distribution histogram was also generated to understand how the angular errors were spread across the selected sample.

2.6 Saving the Results

The generated graphs and tables were saved in the Week5_Results folder in Google Drive.

The saved outputs include the class-wise error table, orientation-range analysis table, highest-error prediction table, graphs, and a text summary of the numerical observations.

This organisation allows the results to be reviewed and reused in subsequent experiments without repeating the initial dataset-processing steps.

3. Results and Observations

3.1 Baseline Comparison from Week 4

The Week 4 experiment established the baseline performance of the three methods on the same 100 vehicle samples.

Table 1. Baseline Comparison of Orientation-Estimation Methods

Method	**Successful Predictions**	**Failed Predictions**	**Mean Angular Error**	**Median Angular Error**	**Standard Deviation**	**Maximum Error**
PCA	100	0	10.74°	4.85°	16.11°	81.71°
Hough Transform	98	2	12.46°	4.18°	20.44°	82.39°
Contour-Based	100	0	10.97°	5.72°	16.44°	89.00°

The baseline results showed that Hough Transform had the highest mean angular error and standard deviation among the three methods. It also failed to produce valid predictions for two samples.

But the median error for Hough Transform was the lowest at 4.18°. This indicates that its typical error was less than its mean error would suggest, and a smaller number of cases with large errors may have had a significant impact on its mean and variance.

These results motivated a more detailed investigation of the Hough Transform rather than an immediate conclusion that it performs poorly on every vehicle.

3.2 Class-Wise Hough Transform Performance

The class-wise analysis was performed using valid Hough Transform predictions from the selected evaluation sample. For each vehicle class, the sample count, mean angular error, median angular error, maximum angular error, and standard deviation were calculated.

Table 2 — Class-wise statistical analysis

Vehicle_Class	Sample_Count	Mean_Error_deg	Median_Error_deg	Maximum_Error_deg	Standard_Deviation_deg
Container	2	49.129999999999995	49.129999999999995	77.16	39.64040615331785
Auto3WCargo	3	38.583333333333336	19.979999999999997	81.49	37.26739906853352
Trailer	3	32.866666666666667	8.909999999999997	82.39	42.896018851792455
PickUp	4	20.7175	8.059999999999999	65.27	30.115037633492452
Truck	5	17.502000000000002	4.930000000000007	61.81	25.33429138539304
AutoRicksaw	8	13.96875	8.07	63.49	20.59663733823697
SUV	34	12.973235294117648	4.055	79.96	21.044624840329934
Sedan	7	7.032857142857144	3.8	19.86	7.709197351833964
MotorCycle	29	4.543448275862069	2.79	18.350000000000001	4.095297194746738
Tanker	1	3.2699999999999996	3.2699999999999996	3.2699999999999996	0.0

Bus	2	0.7349999999999994	0.7349999999999994	1.0100000000000051	0.38890872965260914

In the Week 5 class-wise analysis, Container had the highest mean angular error at 49.13°, based on two valid predictions. Auto3WCargo and Trailer followed, with mean errors of 38.58° and 32.87°, respectively, based on three valid predictions each. MotorCycle had a mean error of 4.54° across 29 valid predictions, while Bus had a mean error of 0.73° across two valid predictions.

These results show that the observed angular error varied across vehicle classes in the selected evaluation sample. Differences in vehicle geometry, edge structure, and the number of prominent lines may affect the orientation estimated by the Hough Transform.

However, Container had only two samples, while Auto3WCargo and Trailer had three samples each. Therefore, their high mean errors should be considered preliminary observations rather than definitive conclusions about the overall performance of these vehicle classes.

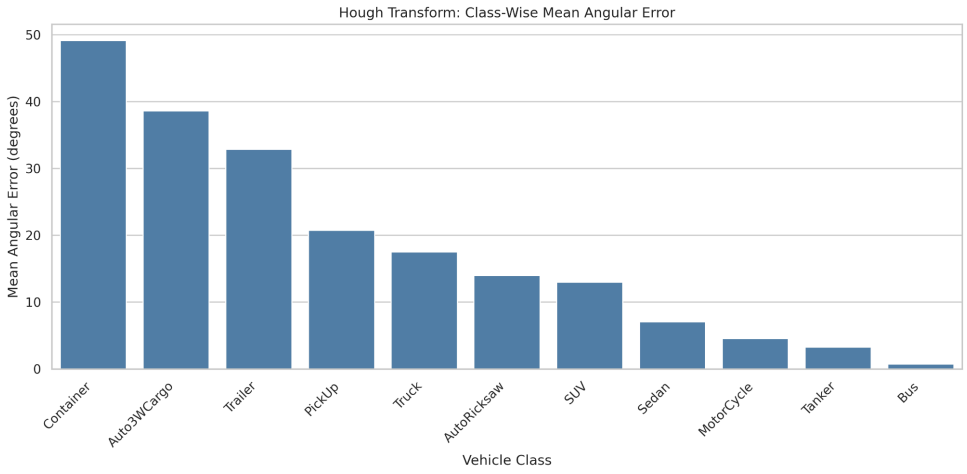


Figure 1: Hough Transform Class-Wise Mean Angular Error

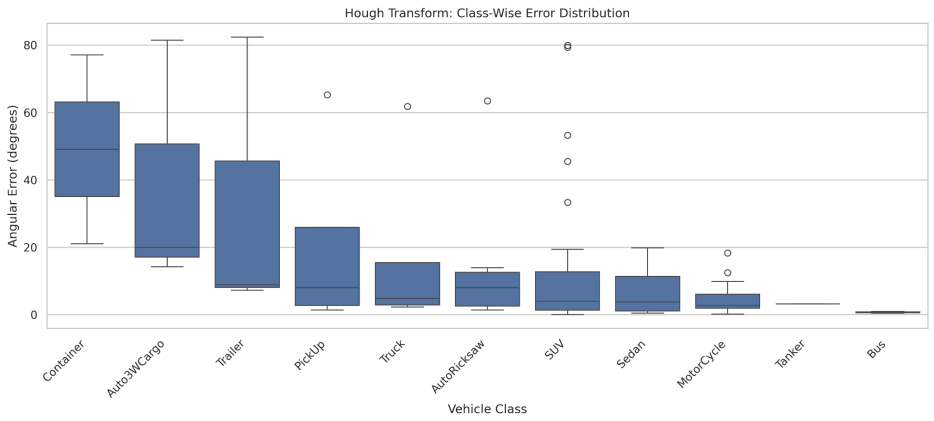


Figure 2: Hough Transform Class-Wise Error Distribution

3.3 Ground-Truth versus Predicted Orientation

The ground-truth versus predicted orientation plot was generated to show how closely the Hough estimates followed the reference orientations.

The diagonal reference line represents ideal agreement. Points away from this line indicate estimation errors, with larger deviations corresponding to larger angular differences.

This plot complements the numerical analysis by showing the individual prediction behaviour across the 0°–90° orientation range.

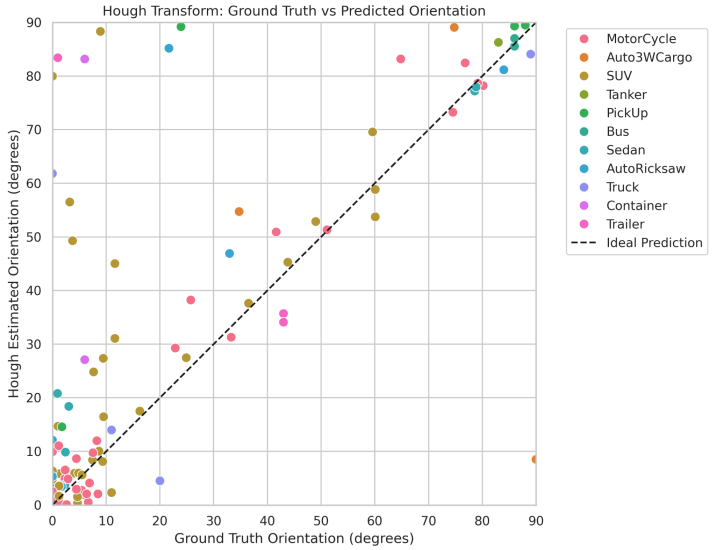


Figure 3: Hough Transform Ground-Truth versus Predicted Orientation

3.4 Error Variation with Vehicle Orientation

The relationship between ground-truth orientation and angular error was examined using a scatter plot and an orientation-range analysis.

The scatter plot shows the error of individual samples across the available orientation range. The grouped analysis compares the mean error for the six orientation intervals.

These analyses were used to investigate whether higher errors were concentrated within particular orientation ranges. Any such pattern must be interpreted in relation to the sample count in each interval.

Table 3. Orientation-Range-Wise Hough Error

Orientation_Range	Sample_Count	Mean_Error_deg
0-15	62	13.320645161290322
15-30	7	23.851428571428574
30-45	8	7.996249999999998
45-60	3	4.689999999999998
60-75	5	8.304
75-90	13	8.396923076923077

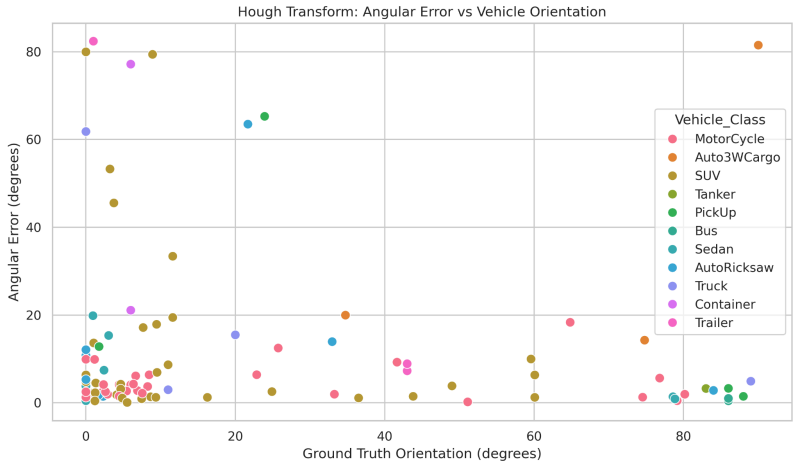


Figure 4: Hough Transform Error versus Ground-Truth Orientation

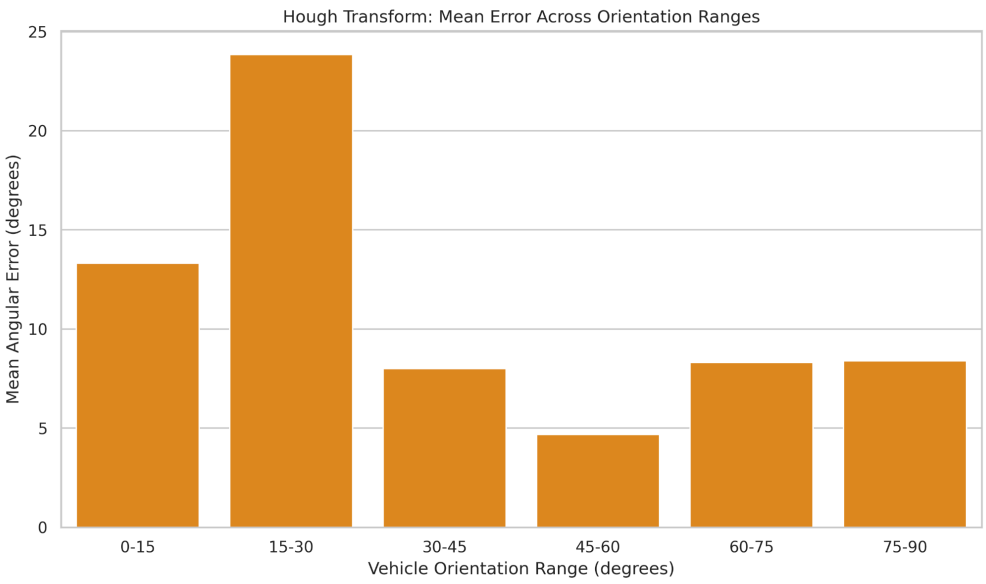


Figure 5: Hough Transform Mean Error Across Orientation Ranges

3.5 Distribution of Angular Errors and High-Error Samples

The error histogram was created to investigate the distribution of the angular errors of the Hough Transform. This will help to see if errors are clustered at low values, or if there are many large errors.

The highest-error prediction table was also created to identify individual samples that require further inspection.

The five highest-error samples should be examined in detail before drawing conclusions about the physical reasons for the estimation failures.

Table 4. Five Highest-Error Hough Predictions

Vehicle_ID	Vehicle_Class	Ground_Truth_Orientation_deg	Hough_Estimated_Orientation_deg	Hough_Angular_Error_deg
cLEH_vQvJ_zAUO_vehicle_1	Trailer	1.0	83.39	82.39
hwun_OIDq_ey5G_vehicle_1	Auto3WCargo	90.0	8.51	81.49

dH2R_2Thg_72eh_vehicle_1	SUV	0.0	79.96	79.96
wmtT_f6a4_vb0t_vehicle_1	SUV	8.92	88.31	79.39
y0bo_oHBY_y2mz_vehicle_1	Container	6.0	83.16	77.16

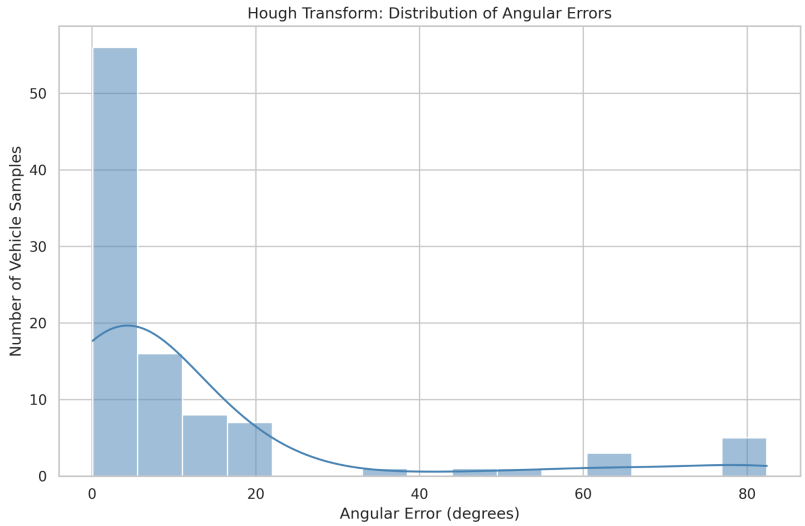


Figure 6: Hough Transform Angular Error Distribution

3.6 Possible Reasons for High Hough Transform Error

The Hough Transform estimates orientation by detecting line segments in the edge image and combining their directions. However, the detected lines do not necessarily correspond to the longitudinal axis of the vehicle.

The following factors may contribute to the observed errors:

1. **Multiple prominent edges:** Roof boundaries, windows, and other vehicle structures may produce lines at different angles. The combined direction may differ from the vehicle's main axis.
2. **Background interference:** Road markings, shadows, and other structures within the crop may introduce irrelevant line segments.

3. **Vehicle geometry:** Different classes have different shapes and edge patterns. Some vehicles may not contain a clear set of long edges aligned with the ground-truth orientation.
4. **Weak or fragmented edges:** Blurring, low contrast, and incomplete boundaries can affect line detection and reduce the reliability of the estimated angle.
5. **Line-selection and averaging:** The implementation is a combination of detected line directions with line-length weighting. If the irrelevant or misoriented segments are given significant weight, the resulting direction will be in error.
6. **Parameter sensitivity:** Fixed Hough parameters may not work equally well for every crop size or edge structure, potentially contributing to unstable or missing estimates.

These factors are possible explanations based on how the method operates. The numerical error graphs identify where Hough performs poorly, but the specific causes have not yet been confirmed through direct inspection of the detected line segments.

4. Challenges Faced

The main challenge during Week 5 was understanding why Hough Transform had the highest mean angular error despite having the lowest median error in the baseline experiment.

This difference indicates that the method's performance cannot be understood using a single metric. Its error distribution and extreme cases must also be examined.

Another challenge was interpreting the class-wise results. Container, Auto3WCargo, and Trailer showed high mean angular errors in the selected sample. However, Container had only two valid predictions, while Auto3WCargo and Trailer had three each. Therefore, these observations are preliminary and may not represent the performance of Hough Transform across these vehicle classes more generally.

The relationship between ground-truth orientation and angular error also requires careful interpretation. Differences in mean error across orientation ranges do not establish causation, particularly because some ranges contain only a small number of samples.

Finally, numerical analysis alone cannot establish which detected lines caused a high error. Visual inspection of the vehicle crops and their detected Hough line segments is needed to determine whether the incorrect orientation resulted from background interference, vehicle geometry, fragmented edges, or the line-combination procedure.

5. Current Status

The Week 5 quantitative analysis focuses on Hough Transform using the existing 100-vehicle evaluation dataset.

The work includes:

- Loading and validating the Week 3 result table.
- Recalculating Hough angular errors for valid predictions.
- Generating class-wise error statistics.
- Identifying classes with high mean angular errors.
- Comparing ground-truth and predicted orientations.
- Analysing angular errors across orientation ranges.
- Identifying individual samples with the largest errors.
- Generating graphs and statistical tables.
- Saving the outputs in the Week5_Results folder in Google Drive.

The Week 4 baseline showed that Hough Transform had a mean angular error of 12.46°, a standard deviation of 20.44°, and two failed predictions. In the Week 5 class-wise analysis of valid predictions, Container had the highest mean angular error at 49.13°, based on two valid samples. These findings provide a starting point for investigating the limitations of Hough Transform, but they do not confirm the specific image-level causes of the errors.

6. Work to Be Done Next Week

In the next stage, we plan to:

- Inspect the vehicle crops with the five highest Hough errors.
- Visualise the corresponding edge maps and detected Hough line segments.

- Identify whether the detected lines align with the vehicle's longitudinal axis.
- Investigate the effects of background edges and different vehicle geometries.
- Test suitable adjustments to Hough parameters, including line-detection threshold, minimum line length, and maximum line gap.
- Compare the revised Hough predictions with the current baseline using the same vehicle samples.
- Document any improvements and remaining failure cases.

The next phase will focus on validating the possible causes of high Hough error and testing targeted improvements rather than repeating the full three-method analysis.

7. References

[1] Y. Bhavsar, M. Zaveri, M. Raval, K. R. Patel, and S. B. Zaveri, "Descriptor: Drone Nadir-view Annotated Images of Vehicles Detection Dataset for India with Heading-angle Oriented Bounding Box (DRASHTI-HaOBB)," *IEEE Data Descriptions*, 2026. Available: https://ieeexplore.ieee.org/document/11421776.

[2] Y. Bhavsar, M. Zaveri, M. Raval, S. Zaveri, and Ahmedabad University, "DRASHTI-HaOBB: Drone nadir-view Annotated Images of Vehicles Dataset for India - Heading-angle Oriented Bounding Box," *Zenodo*, 2026. Available: https://zenodo.org/records/18278989.